

Configure ETL Pipelines With Llama 3.1 For Data Cleansing

Learn how to integrate the Llama 3.1 8B model into your ETL pipeline for efficient batch inferencing and data cleansing. This guide demonstrates how to switch from default models, execute full table refreshes, and monitor data quality through expectation flows.

1 Click "Jobs & Pipelines"

The screenshot shows the Databricks workspace interface. On the left sidebar, the 'Jobs & Pipelines' icon is highlighted with an orange circle. The main content area displays the workspace for user 'visheshh.arya87@gmail.com' under the name 'bharatbricksiitb'. A table lists various items in the workspace:

Name	Type	Owner	Created at
.databricks	Folder	user_9501cd52	Apr 10, 2026, 08...
02-data-transformation	Folder	user_9501cd52	Apr 10, 2026, 08...
05-iitb-junta-analytics-genie	Folder	user_9501cd52	Apr 10, 2026, 08...
06-iitb-baap-agent	Folder	user_9501cd52	Apr 10, 2026, 08...
instructions	Folder	user_9501cd52	Apr 10, 2026, 08...
raw_data	Folder	user_9501cd52	Apr 10, 2026, 08...
.gitignore	File	user_9501cd52	Apr 10, 2026, 08...
01-data-ingestion	Notebook	user_9501cd52	Apr 10, 2026, 08...
03-metric-view	Notebook	user_9501cd52	Apr 10, 2026, 08...
04-life-at-iit-bombay	Dashboard	user_9501cd52	Apr 10, 2026, 08...
README.md	File	user_9501cd52	Apr 10, 2026, 08...

2 Click "ETL pipeline"

The screenshot shows the Databricks 'Jobs & Pipelines' page. On the left is a sidebar with navigation options like Home, Workspace, Recents, Catalog, and Jobs & Pipelines (which is selected). The main content area has a 'Create new' section with three cards: 'Ingestion pipeline', 'ETL pipeline' (highlighted with an orange circle), and 'Job'. Below this is a table titled 'Jobs & pipelines' with columns: Name, Type, Tags, Run as, Trigger, and Recent runs. The table is currently empty, showing a message 'No jobs or pipelines found.' A notification at the top right indicates 'Deleting pipeline "02-data-transformation".'

3 Type "02-data-transformation"

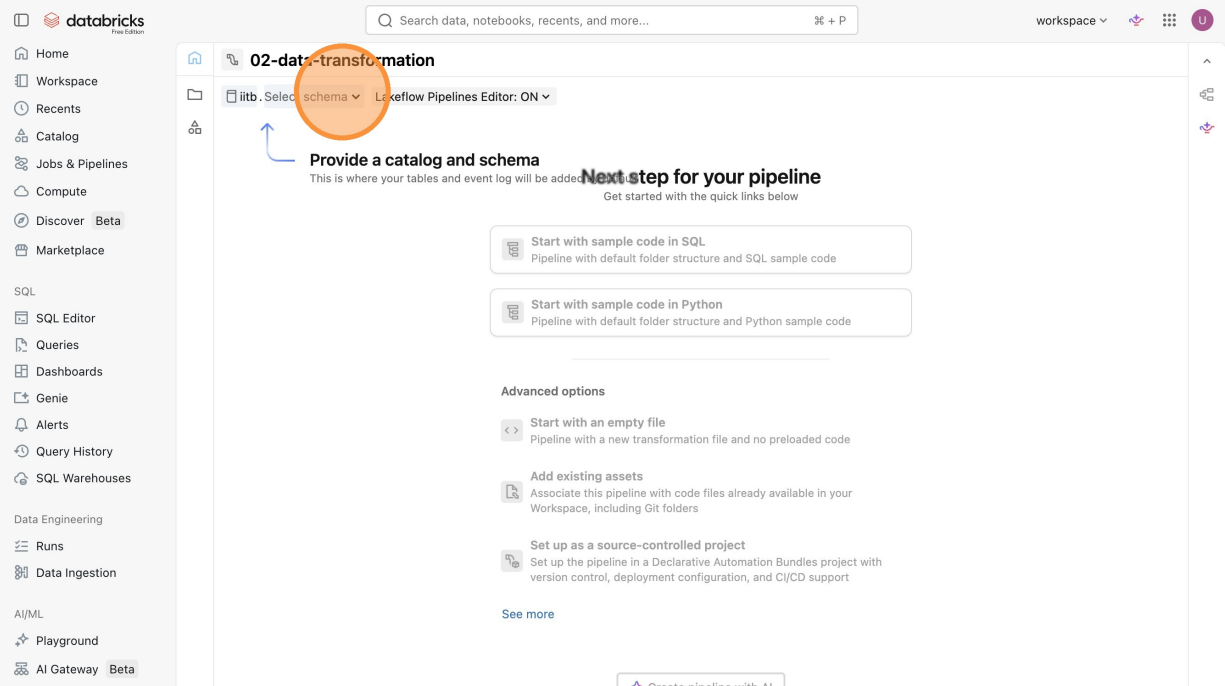
4 Click "workspace"

The screenshot shows the Databricks workspace interface. On the left sidebar, the 'workspace' button is highlighted with an orange circle. The main panel displays the '02-data-transformation' pipeline editor. The top bar shows 'workspace' and 'Lakeflow Pipelines Editor: ON'. The main content area has a heading 'Give your pipeline a name' and a subheading 'Next step for your pipeline'. Below this, there are four options to start a new pipeline: 'Start with sample code in SQL', 'Start with sample code in Python', 'Start with an empty file', and 'Add existing assets'. The 'Start with an empty file' option is selected. The 'workspace' button in the left sidebar is highlighted with an orange circle.

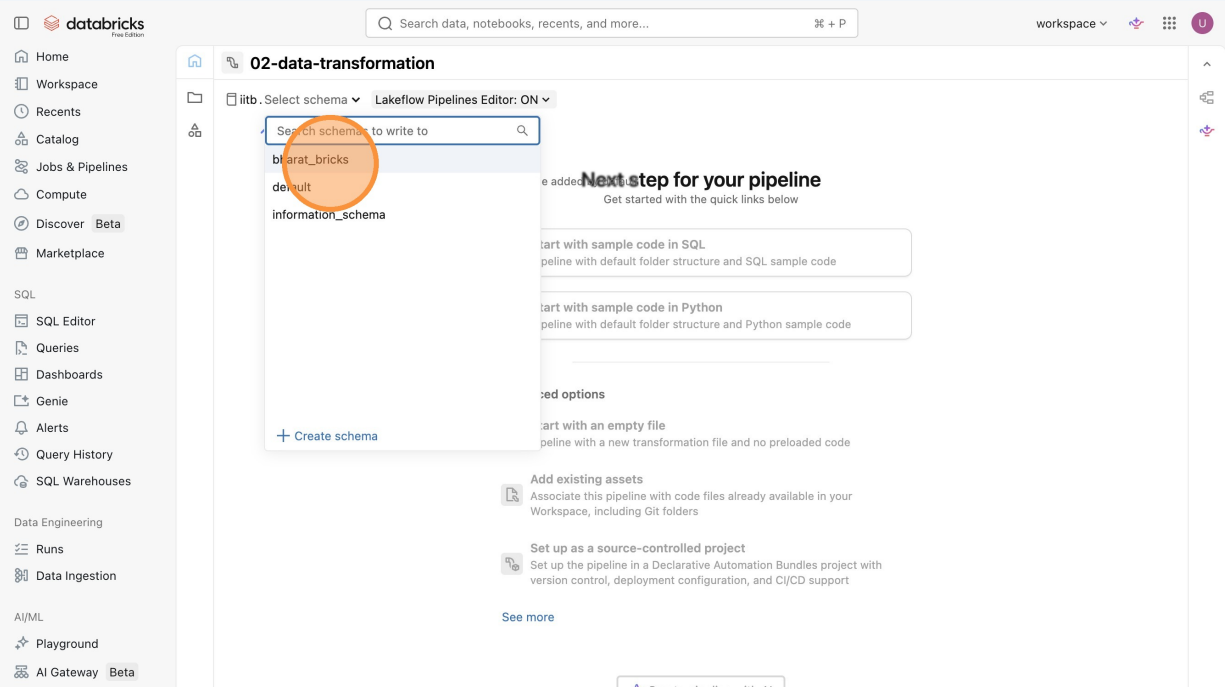
5 Click "iitb"

The screenshot shows the Databricks workspace interface. On the left sidebar, the 'iitb' button is highlighted with an orange circle. The main panel displays the '02-data-transformation' pipeline editor. The top bar shows 'workspace' and 'Lakeflow Pipelines Editor: ON'. The main content area has a heading 'Give your pipeline a name' and a subheading 'Next step for your pipeline'. Below this, there are four options to start a new pipeline: 'Start with sample code in SQL', 'Start with sample code in Python', 'Start with an empty file', and 'Add existing assets'. The 'Start with an empty file' option is selected. The 'iitb' button in the left sidebar is highlighted with an orange circle.

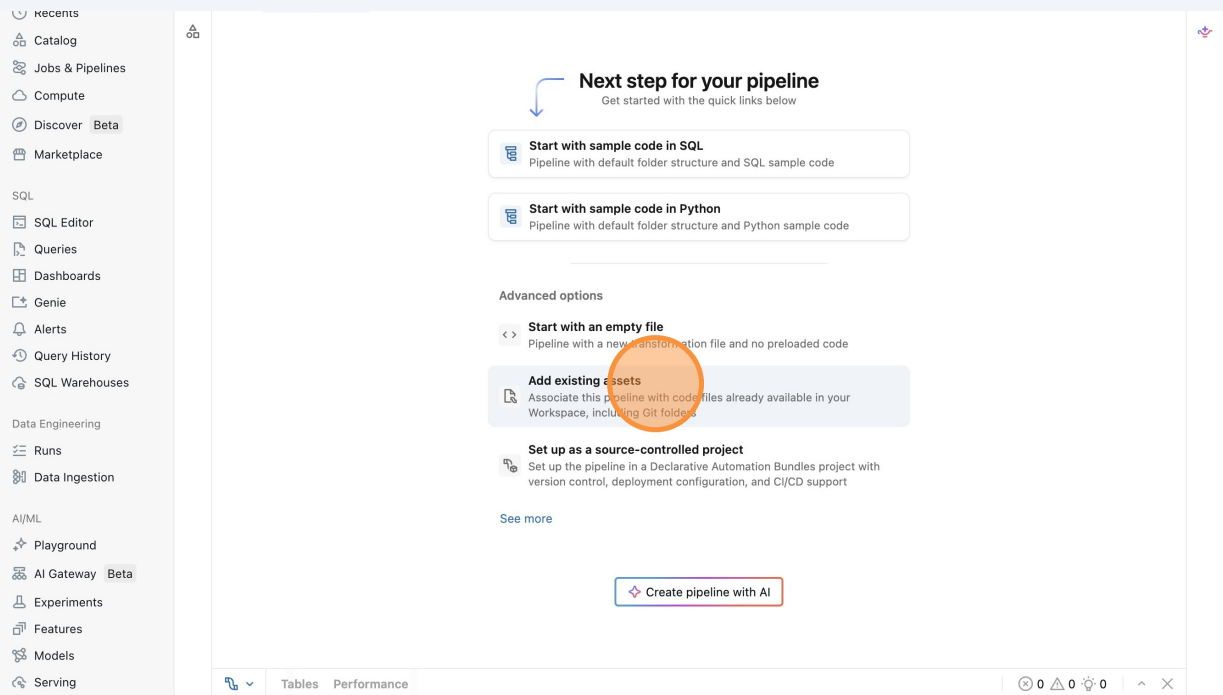
6 Click "Select schema"



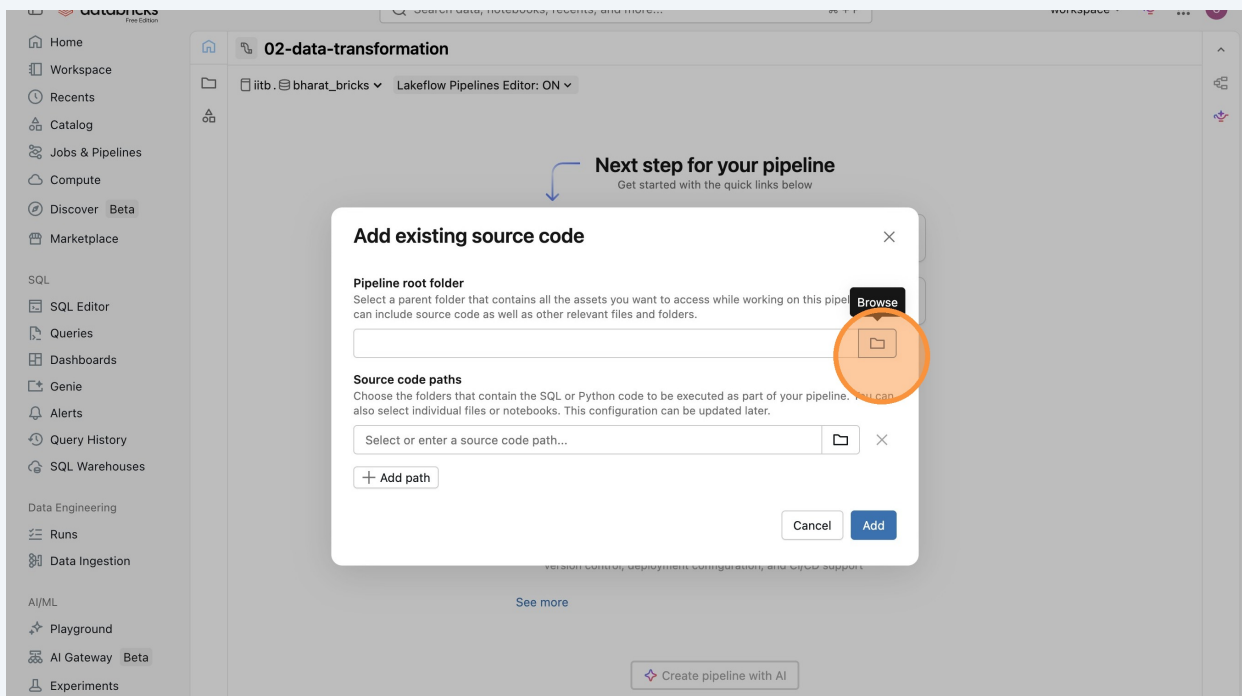
7 Click "bharat_bricks"



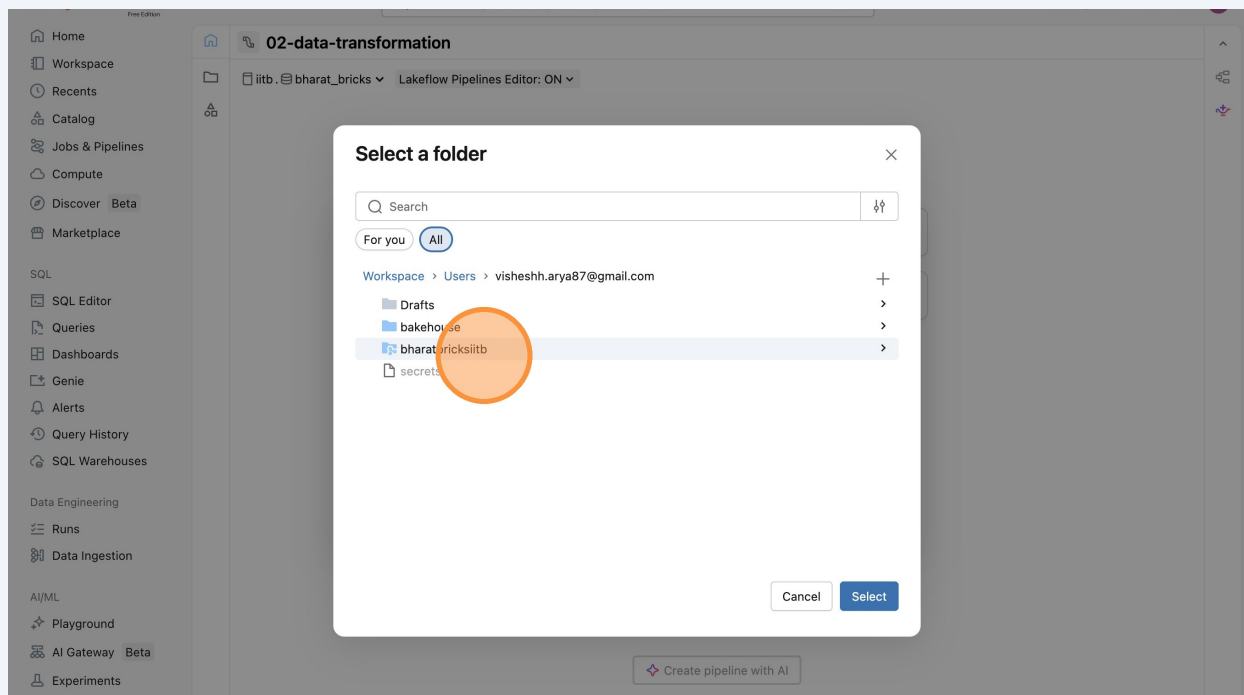
8 Click "Add existing assets"



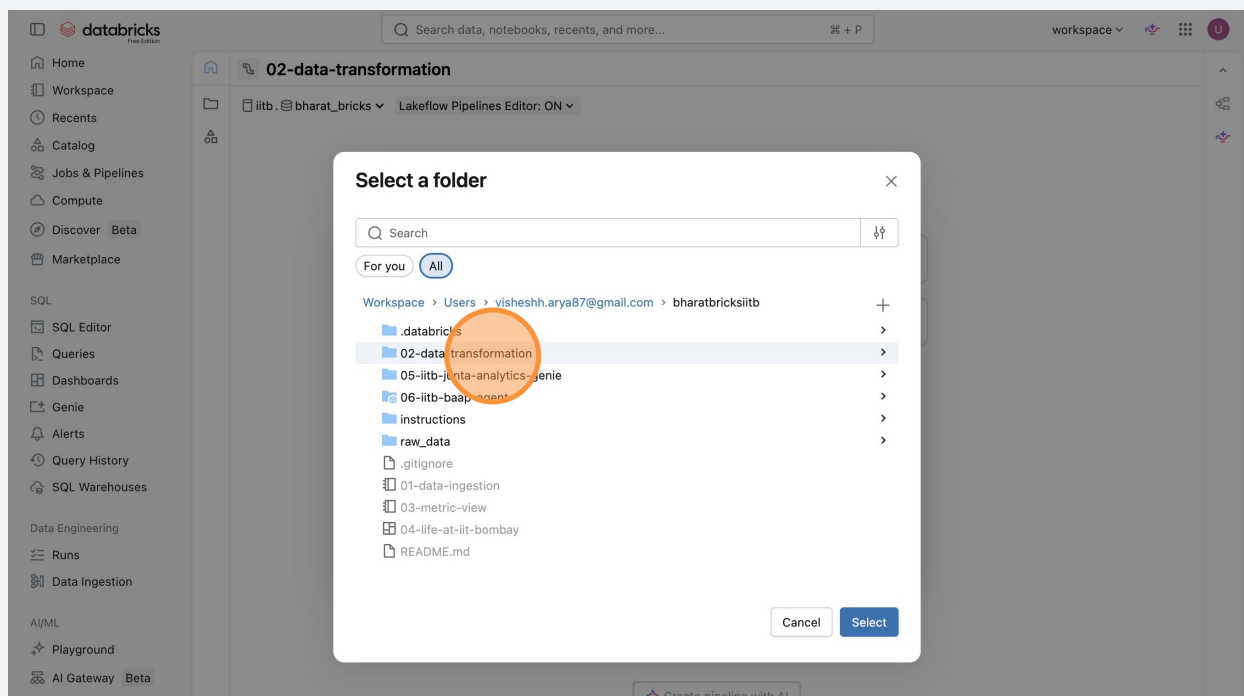
9 Click "loading"



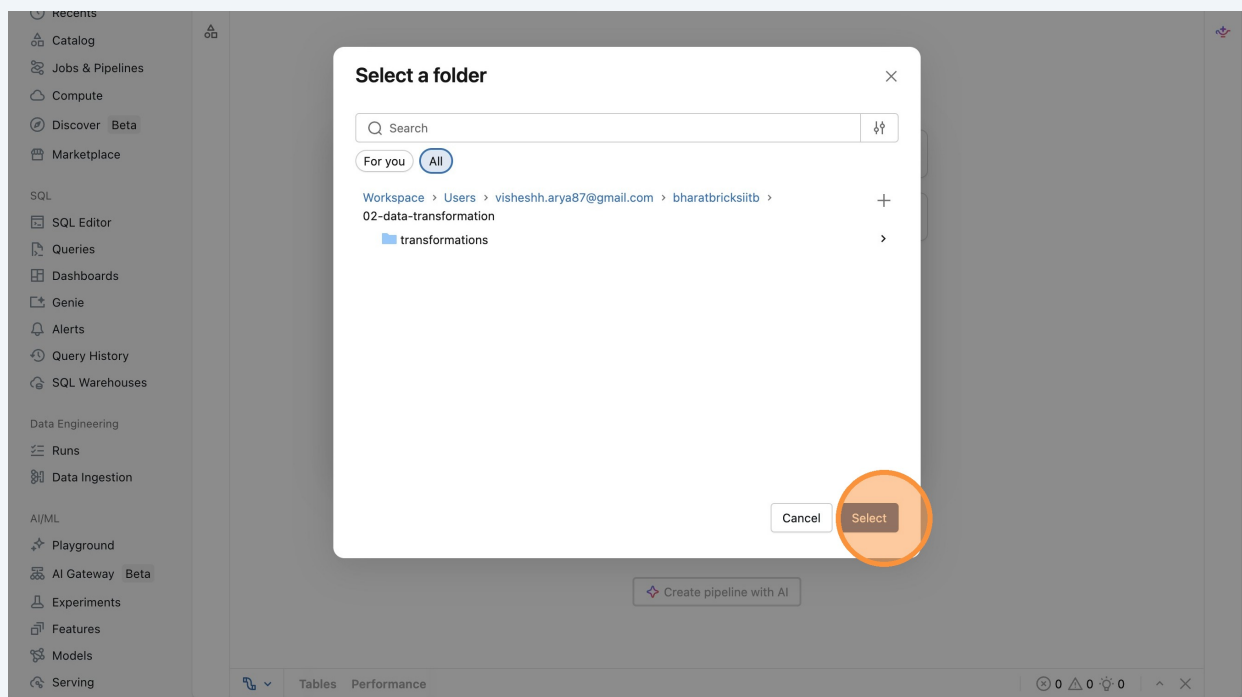
10 Click "bharatbricksiitb"



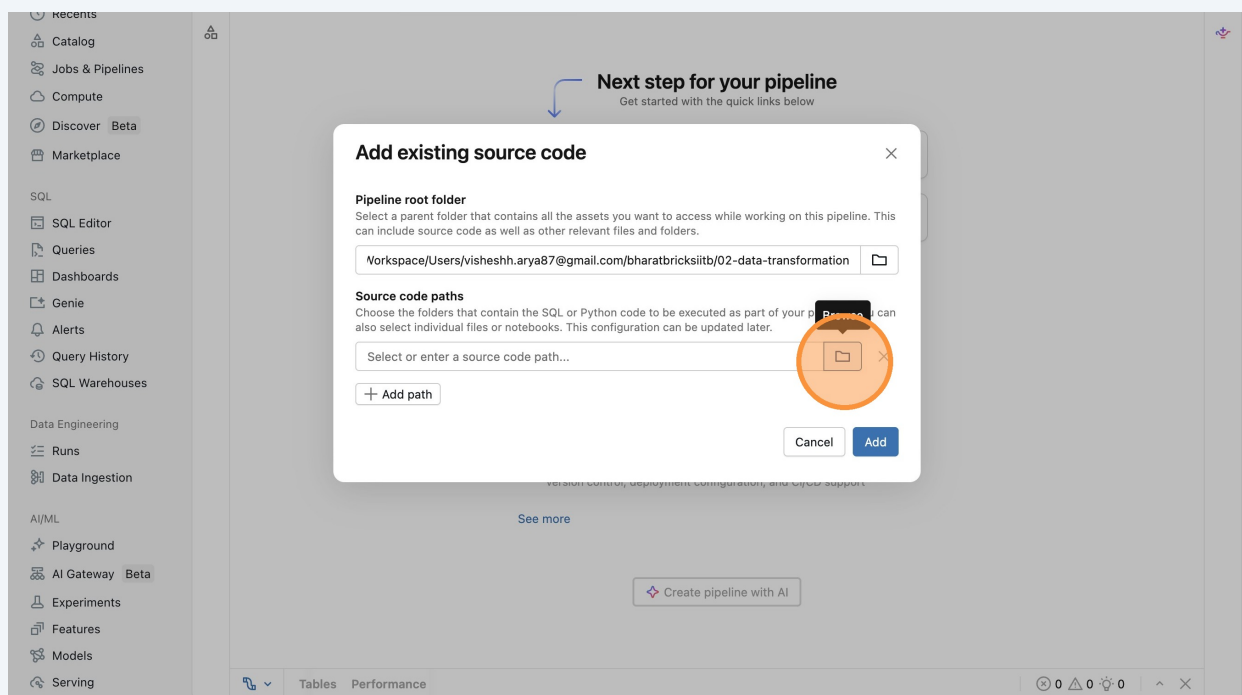
11 Click "02-data-transformation"



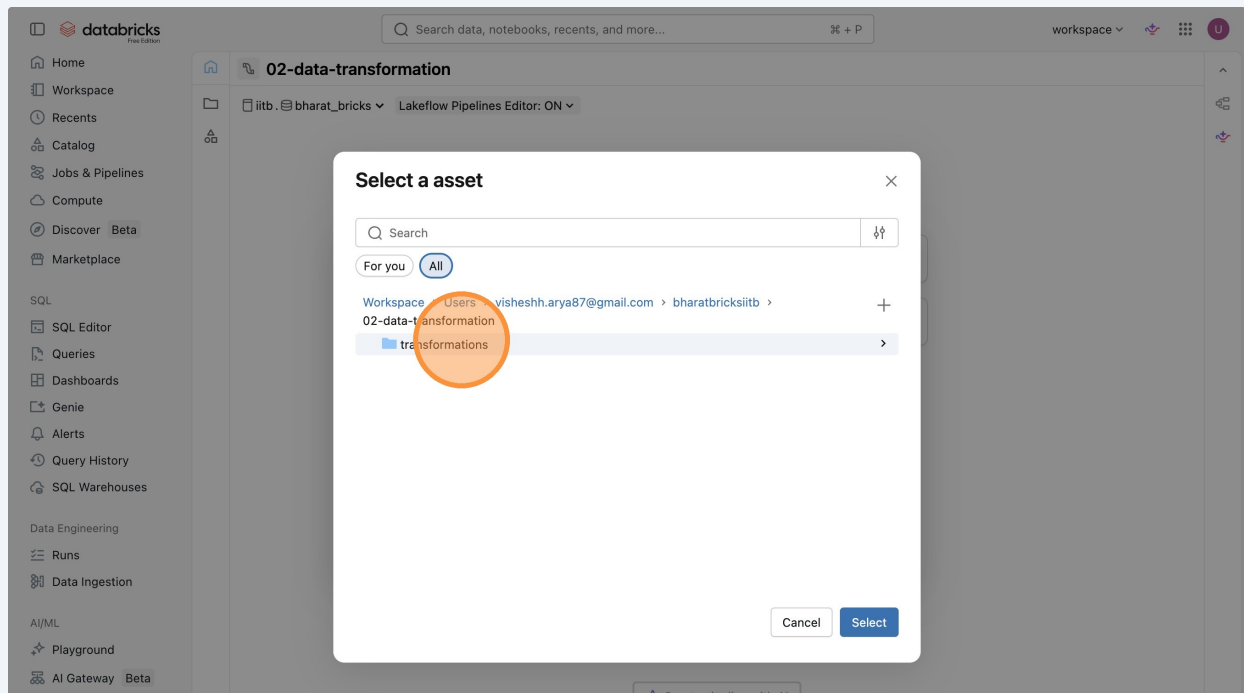
12 Click "Select"



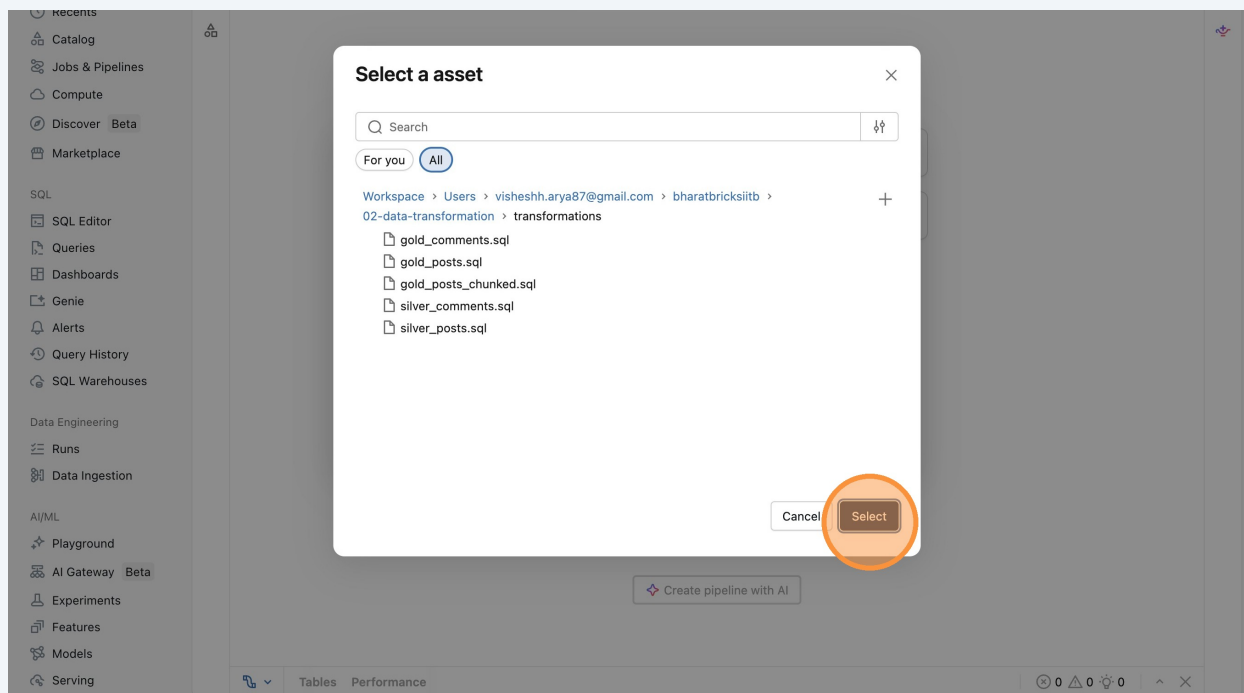
13 Click here.



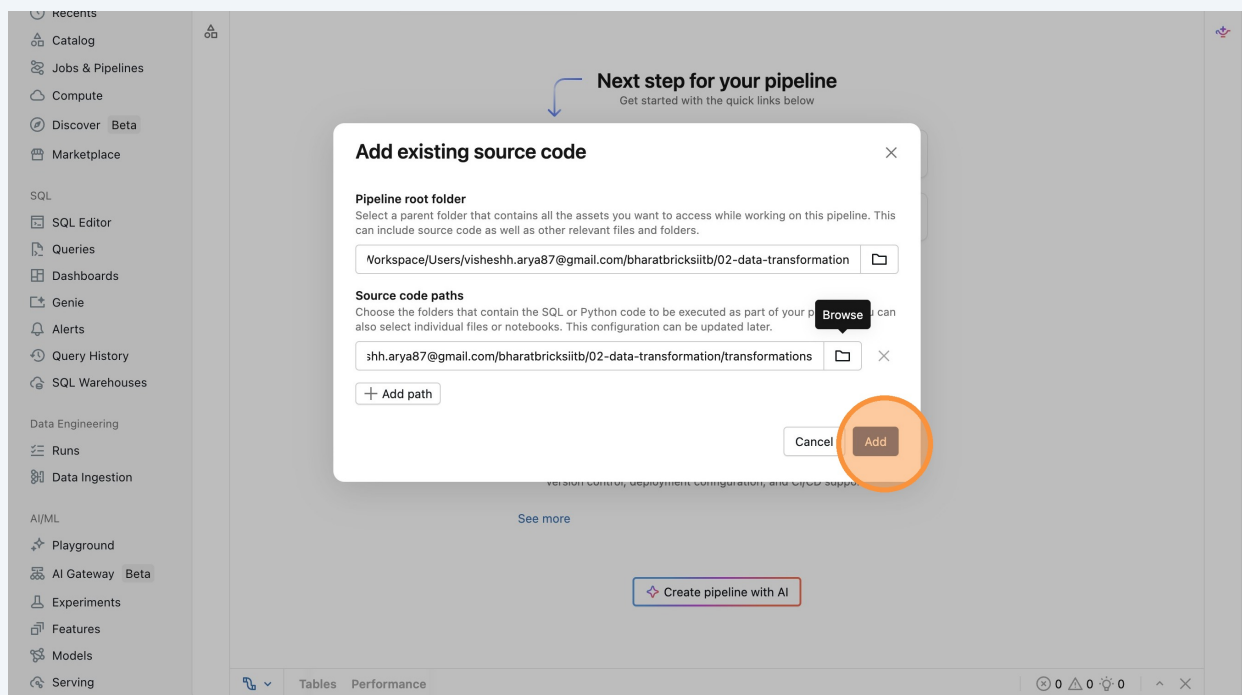
14 Click "transformations"



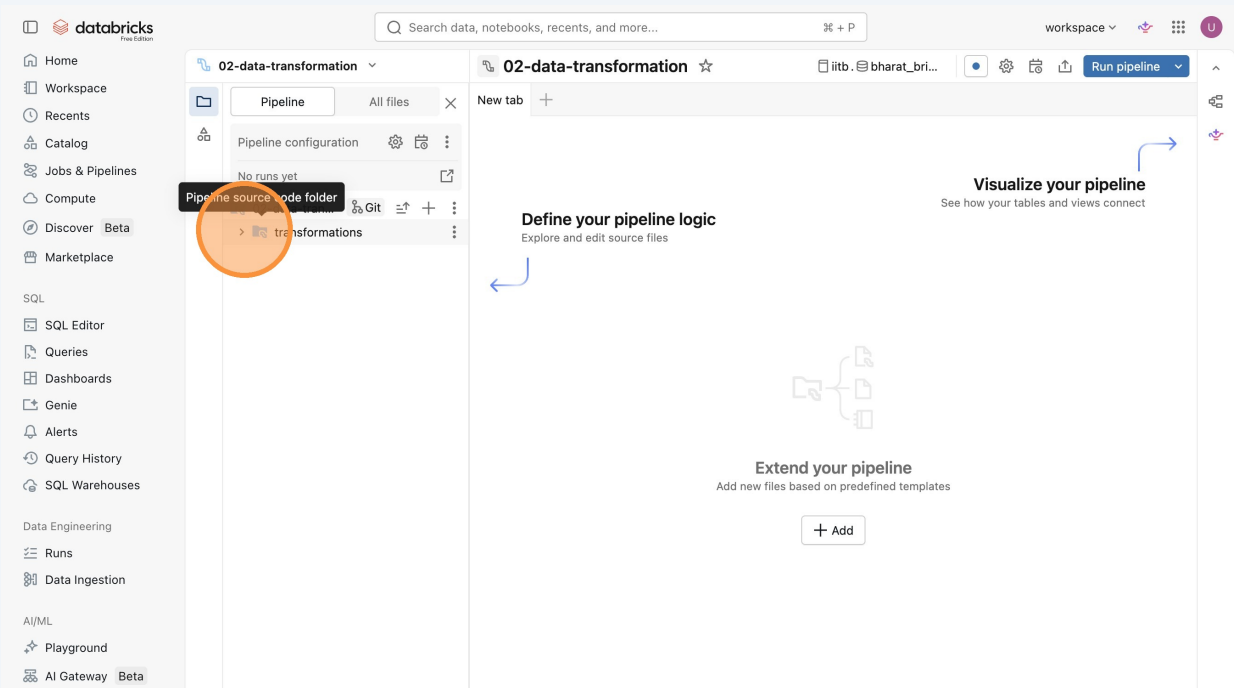
15 Click "Select"



16 Click "Add"



17 Click this icon.



18 Click "silver_posts.sql"

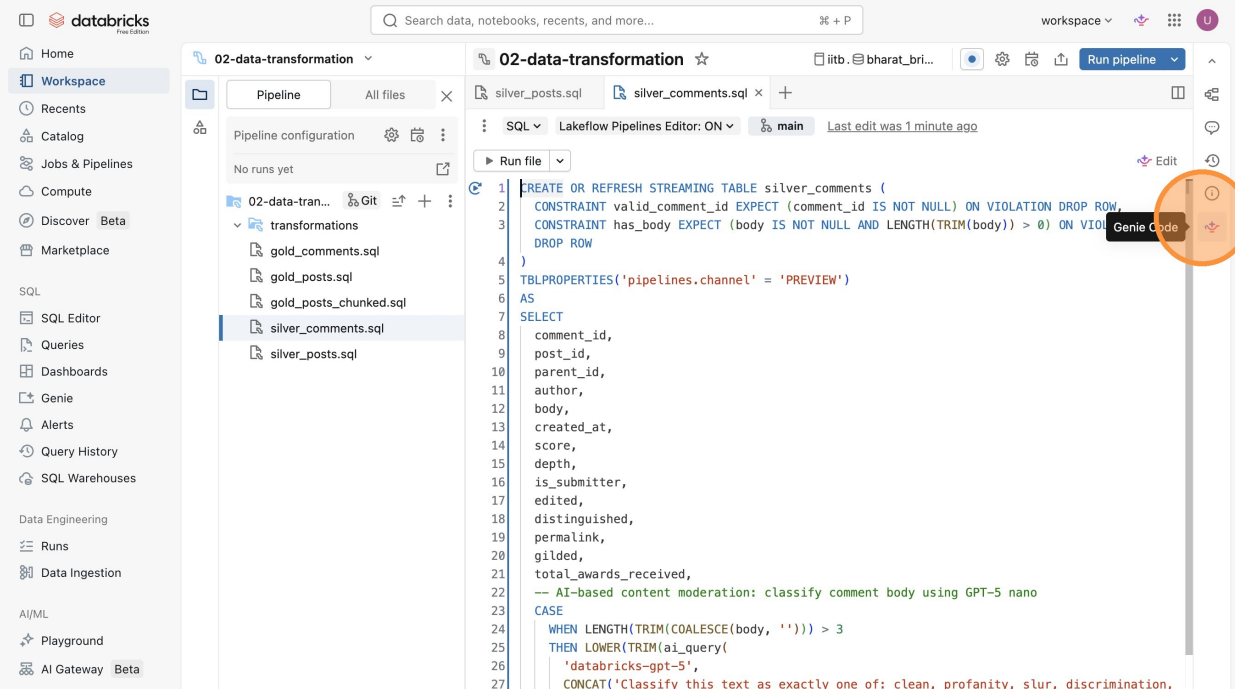
The screenshot shows the Databricks workspace interface. On the left sidebar, the 'Workspace' tab is selected. The main area displays the '02-data-transformation' pipeline configuration. The 'transformations' list on the left contains the following files: 'gold_comments.sql', 'gold_posts.sql', 'gold_posts_chunked.sql', 'silver_comments.sql', and 'silver_posts.sql'. The 'silver_posts.sql' file is highlighted with an orange circle. The main editor area shows a 'Define your pipeline logic' section with a 'Visualize your pipeline' button and an 'Extend your pipeline' section with an 'Add' button.

19 Click "silver_comments.sql"

The screenshot shows the Databricks workspace interface. On the left sidebar, the 'Workspace' tab is selected. The main area displays the '02-data-transformation' pipeline configuration. The 'transformations' list on the left contains the following files: 'gold_comments.sql', 'gold_posts.sql', 'gold_posts_chunked.sql', 'silver_comments.sql', and 'silver_posts.sql'. The 'silver_comments.sql' file is highlighted with an orange circle. The main editor area shows the SQL code for the 'silver_posts' table, including a 'CREATE OR REFRESH STREAMING TABLE' statement and a 'SELECT' query.

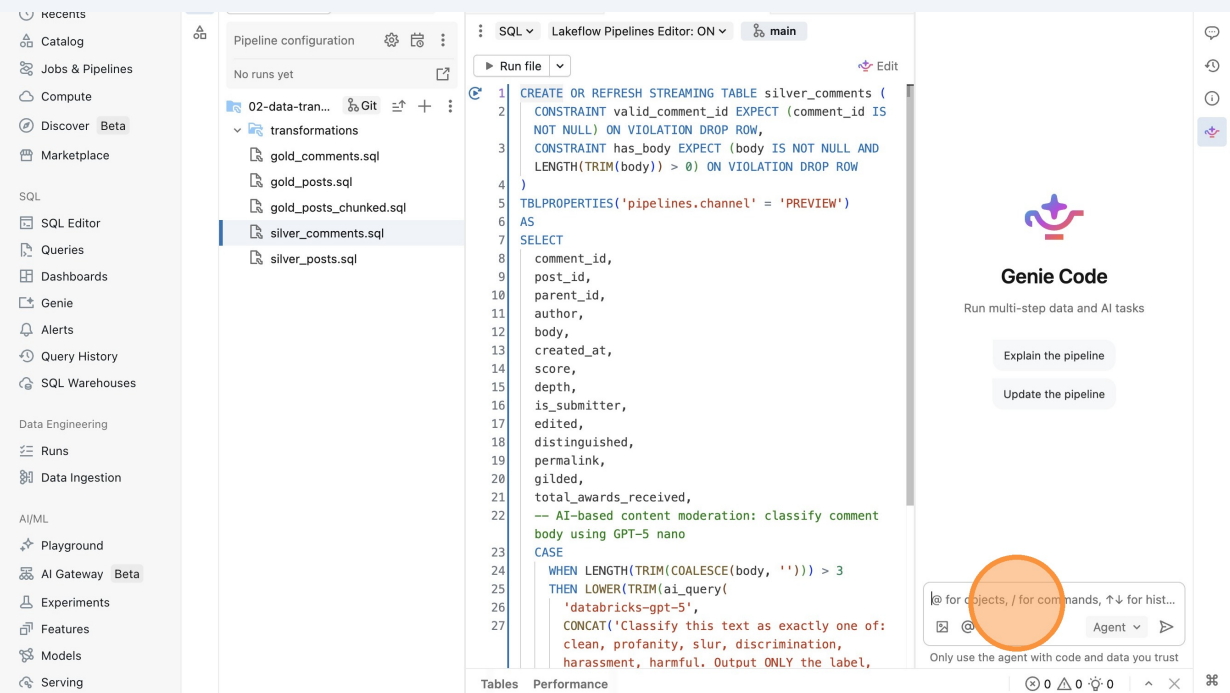
```
1 CREATE OR REFRESH STREAMING TABLE silver_posts (  
2   CONSTRAINT valid_post_id EXPECT (post_id IS NOT NULL) ON VIOLATION DROP ROW,  
3   CONSTRAINT has_title EXPECT (title IS NOT NULL AND LENGTH(TRIM(title)) > 0) ON VIOLATION  
4   DROP ROW  
5 )  
6 TBLPROPERTIES('pipelines.channel' = 'PREVIEW')  
7 AS  
8 SELECT  
9   post_id,  
10  title,  
11  body,  
12  author,  
13  created_at,  
14  permalink,  
15  score,  
16  upvote_ratio,  
17  num_comments,  
18  flair,  
19  author_flair_text,  
20  is_self,  
21  is_video,  
22  is_original_content,  
23  is_nsfw,  
24  spoiler,  
25  locked,  
26  domain,  
27  num crossposts.
```

20 Click "Genie Code"



The screenshot shows the Databricks workspace interface. On the left is a sidebar with navigation options like Home, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, Playground, and AI Gateway (Beta). The main area is titled '02-data-transformation' and shows a pipeline configuration on the left and a SQL query in the center. The query is for creating or refreshing a streaming table named 'silver_comments' with various constraints and a SELECT statement. A red circle highlights the 'Genie Code' button in the top right corner of the editor.

21 Click here.



The screenshot shows the Databricks workspace interface. On the left is a sidebar with navigation options like Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, Playground, AI Gateway (Beta), Experiments, Features, Models, and Serving. The main area is titled '02-data-transformation' and shows a pipeline configuration on the left and a SQL query in the center. The query is for creating or refreshing a streaming table named 'silver_comments' with various constraints and a SELECT statement. A red circle highlights the 'Genie Code' button in the top right corner of the editor.

22

Type "i want to use llama 3.1 8b for batch inferencing for cleansing raw data instead of gpt-5 as gpt-5 may not be available in the free edition **Enter**"

23

Click "Ask every time"

The screenshot shows the Lakeflow Pipelines Editor interface. On the left is a sidebar with navigation options like 'recents', 'Catalog', 'Jobs & Pipelines', 'Compute', 'Discover', 'Marketplace', 'SQL', 'SQL Editor', 'Queries', 'Dashboards', 'Genie', 'Alerts', 'Query History', 'SQL Warehouses', 'Data Engineering', 'Runs', 'Data Ingestion', 'AI/ML', 'Playground', 'AI Gateway', 'Experiments', 'Features', 'Models', and 'Serving'.

The main area is divided into three panels:

- Pipeline configuration:** Shows a list of transformations including '02-data-tran...', 'gold_comments.sql', 'gold_posts.sql', 'gold_posts_chunked.sql', 'silver_comments.sql' (highlighted with '+2 -2'), and 'silver_posts.sql'.
- SQL Editor:** Displays a SQL query for creating a streaming table and selecting data. The query includes comments about AI-based content moderation using Llama 3.1 8B. A 'Run file' button is visible at the top.
- Chat Window:** On the right, a chat interface shows a user message: "i want to use llama 3.1 8b for batch inferencing for cleansing raw data instead of gpt-5 as gpt-5 may not be available in the free edition". The system response explains the update and provides a 'Dry-running pipeline' button with a dropdown menu set to 'Ask every time' (circled in orange), along with 'Skip' and 'Allow' buttons.

At the bottom right, there is a section for '1 asset' showing 'silver_comments.sql' with '+2 -2' changes and a 'Send follow-up messages to guide Genie...' button.

24 Click "Always allow in current thread"

The screenshot shows the Databricks Lakeflow Pipelines Editor. On the left, a sidebar lists various tools like SQL Editor, Queries, and Dashboards. The main area displays a SQL query for creating a streaming table and performing content moderation using Llama 3.1 8B. The query includes a CASE statement to classify comment bodies as 'clean', 'profanity', 'slur', 'discrimination', 'harassment', or 'harmful'. On the right, the Genie AI assistant interface is visible, showing a user prompt: "i want to use llama 3.1 8b for batch inferencing for cleansing raw data instead of gpt-5 as gpt-5 may not be available in the free edition". The assistant's response is: "I'll update the code to use Llama 3.1 8B instead of GPT-5 for the content moderation inference. Let me make that change and verify it works." Below the response, there are buttons for "Ask every time", "Always allow in current thread" (highlighted with an orange circle), and "Always allow".

25 Click "Accept all"

The screenshot shows the Databricks workspace. On the left, a sidebar lists various tools like SQL Editor, Queries, and Dashboards. The main area displays a pipeline run for "02-data-transformation". The pipeline configuration shows a SQL job with a query that classifies comment bodies. The Genie AI assistant interface is visible on the right, showing a user prompt: "i want to use llama 3.1 8b for batch inferencing for cleansing raw data instead of gpt-5 as gpt-5 may not be available in the free edition". The assistant's response is: "I'll update the code to use Llama 3.1 8B instead of GPT-5 for the content moderation inference. Let me make that change and verify it works." Below the response, there are buttons for "Run the pipeline update", "Check inference performance", and "Convert to batch UDF in Python". On the right, there are buttons for "Reject all" and "Accept all" (highlighted with an orange circle). Below the Genie interface, a table of results is shown, listing columns like Name, Catalog, Schema, Type, D..., O..., Expect..., Dr..., W..., and Fa... The table contains two rows of data.

Name	Catalog	Schema	Type	D...	O...	Expect...	Dr...	W...	Fa...
gold...	iitb	bharat...	Materia...	-	-	1 defined	-	-	0
gold...	iitb	bharat...	Materia...	-	-	1 defined	-	-	0

26 Click this icon.

The screenshot shows the Databricks workspace interface. On the left is a sidebar with navigation options like Home, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, Playground, and AI Gateway. The main area is titled '02-data-transformation' and shows a pipeline configuration. A 'Run pipeline' button is circled in orange in the top right corner. Below the pipeline configuration, there is a 'Genie Code' section with a text area containing SQL code and a 'Run' button. The bottom section shows a table with 5 columns: Name, Catalog, Schema, Type, D..., O..., Expect..., Dr..., W..., Fa... and 2 rows of data.

27 Click "Run pipeline with full table refresh"

The screenshot shows the Databricks workspace interface, similar to the previous one. The 'Run pipeline' button is circled in orange. Below it, a dropdown menu is visible with the option 'Run pipeline with full table refresh' highlighted. The rest of the interface, including the sidebar, pipeline configuration, and table view, remains the same.

28 Click "Start full refresh"

Full refresh

Full refresh will truncate and recompute ALL tables in this pipeline from scratch. This can lead to data loss for non-idempotent sources. Are you sure you want to full refresh?

Cancel Start full refresh

Tables 5 Performance

Name	Catalog	Schema	Type	D...	O...	Expect...	Dr...	W...	Fa...
gold...	iitb	bharat...	Materia...	-	-	1 defined	-	-	0
gold...	iitb	bharat...	Materia...	-	-	1 defined	-	-	0
gold...	iitb	bharat...	Materia...	-	-	Not defin	-	-	-
silver...	iitb	bharat...	Streami...	-	-	2 defined	-	-	0

29 Click this icon.

02-data-transformation

Open the Pipeline monitoring page in a new browser tab

SQL Lakeflow Pipelines Editor: ON

```
WHEN LENGTH(TRIM(COALESCE(body, ''))) > 3
THEN LOWER(TRIM(ai_query(
  'databricks-meta-llama-3-1-8b-instruct',
  CONCAT('Classify this text as exactly one of:
clean, profanity, slur, discrimination,
harassment, harmful. Output ONLY the label,
nothing else.\nText: ', body)
)))
ELSE 'clean'
END AS content_moderation_label
FROM STREAM(comments)
WHERE
-- Rule-based: remove deleted authors
author != 'deleted'
-- Rule-based: remove deleted/removed comment
bodies
AND body NOT IN ('deleted', 'removed')
AND body IS NOT NULL
```

Setting up tables...

Update type Full refresh all

30 Click "gold_posts"

Jobs & Pipelines

02-data-transformation ☆ New pipeline monitoring: ON [Send feedback](#) [Edit pipeline](#) [Settings](#) [Schedule](#) [Share](#) [Stop](#)

Run: Apr 10, 2026, 08:53 PM · () Running...

Pipeline details

- Pipeline ID: 5e069f67-ee98-4f74-b2e9-4cf1edff...
- Pipeline type: ETL pipeline
- Creator: user_9501cd52
- Owner: user_9501cd52
- Run as: user_9501cd52
- Source code: 1 folder [Open in Editor](#)

Run details

- Pipeline run ID: c2d71cf3-9c34-4547-834b-c6b771...
- Run status: Running...

Tables 8 **Performance** 3

Filter by table Type Status

Name	Catalog	Schema	Type	Durat...	Outp...	Expectations	Drop...	Warn...	Failed
gold_comments	iitb	bharat_bricks	Materialized v...	-	-	1 defined	-	-	0
gold_comments	iitb	bharat_bricks	Materialized v...	-	-	1 defined	-	-	0
gold_posts	iitb	bharat_bricks	Materialized v...	6s	923	1 unmet	-	0	0

31 Click this icon.

Jobs & Pipelines

02-data-transformation ☆ New pipeline monitoring: ON [Send feedback](#) [Edit pipeline](#) [Settings](#) [Schedule](#) [Share](#) [Stop](#)

Run: Apr 10, 2026, 08:53 PM · () Running...

Pipeline details

- Pipeline ID: 5e069f67-ee98-4f74-b2e9-4cf1edff...
- Pipeline type: ETL pipeline
- Creator: user_9501cd52
- Owner: user_9501cd52
- Run as: user_9501cd52
- Source code: 1 folder [Open in Editor](#)

Run details

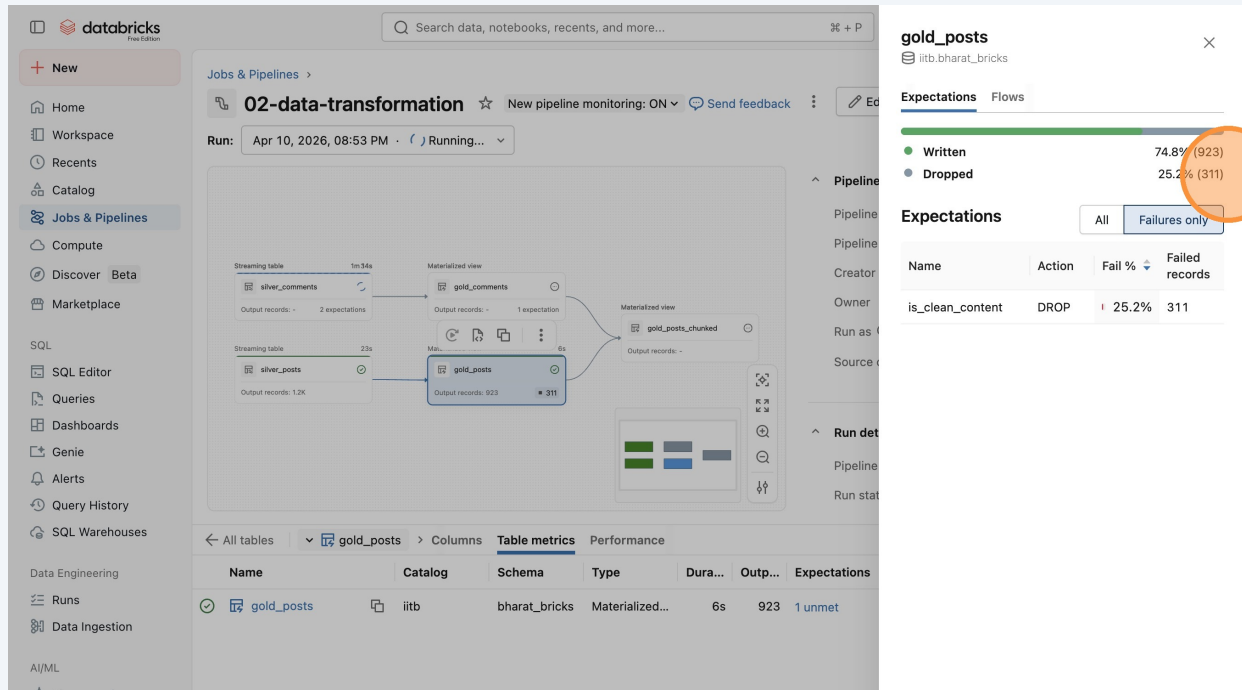
- Pipeline run ID: c2d71cf3-9c34-4547-834b-c6b771...
- Run status: Running...

Table metrics

Name	Catalog	Schema	Type	Dura...	Outp...	Expectations	View data quality	Failed	Incrementa...
gold_posts	iitb	bharat_bricks	Materialized...	6s	923	1 unmet	311	0	Full recom...

32

Click "ExpectationsFlowsWritten74.8% (923)Dropped25.2% (311)ExpectationsAllFailures onlyNameActionFail %Failed recordsis_clean_contentDROP25.2%311"



gold_posts

Expectations

Written 74.8% (923)

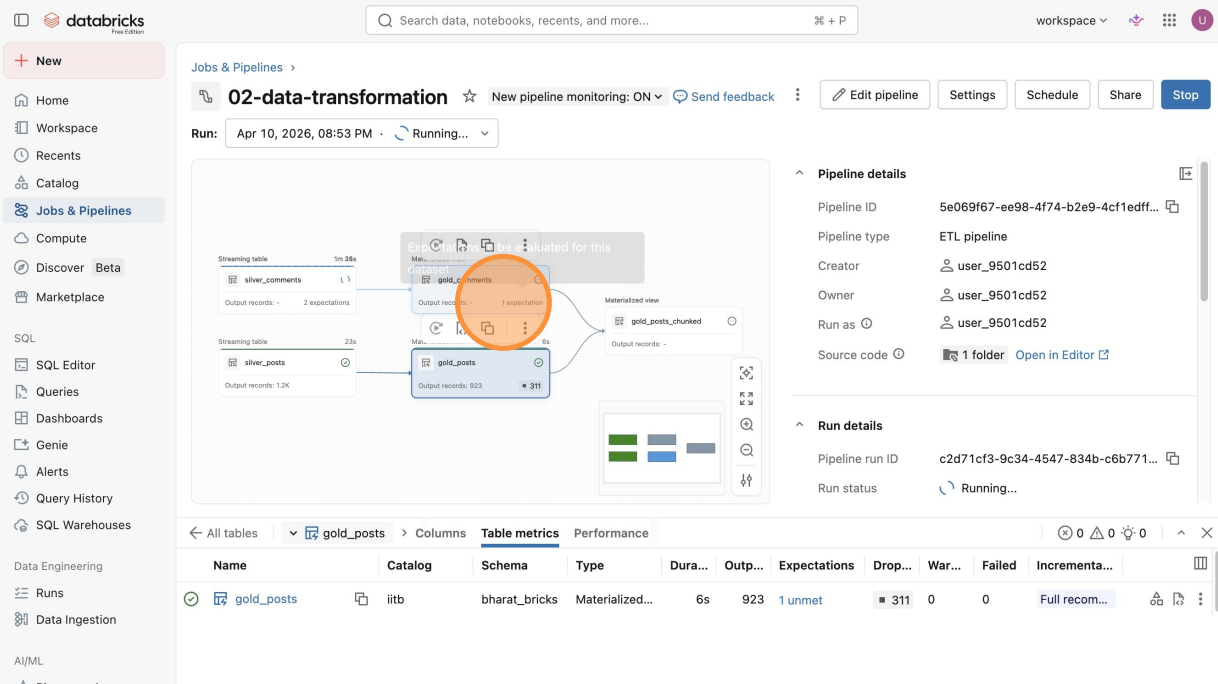
Dropped 25.2% (311)

Expectations All Failures only

Name	Action	Fail %	Failed records
is_clean_content	DROP	25.2%	311

33

Click "1 expectation"



02-data-transformation

Run: Apr 10, 2026, 08:53 PM · Running...

gold_posts

Expectations

Written 74.8% (923)

Dropped 25.2% (311)

Expectations All Failures only

Name	Catalog	Schema	Type	Dura...	Outp...	Expectations	Drop...	War...	Failed	Incrementa...
gold_posts	iitb	bharat_bricks	Materialized...	6s	923	1 unmet	311	0	0	Full recom...

34 Click "5.7K"

Run: Apr 10, 2026, 08:53 PM · () Running...

Run complete

Pipeline details

- Pipeline ID: 5e069f67-ee98-4f74-b2e9-4cf1edff...
- Pipeline type: ETL pipeline
- Creator: user_9501cd52
- Owner: user_9501cd52
- Run as: user_9501cd52
- Source code: 1 folder [Open in Editor](#)

Run details

- Pipeline run ID: c2d71cf3-9c34-4547-834b-c6b771...

Table metrics

Name	Catalog	Schema	Type	Dura...	Outp...	Expectations	Drop...	War...	Failed	Incrementa...
gold_comments	iitb	bharat_bricks	Materialized...	5s	9K	1 unmet	5.7K	0	0	Full recom...

35 Click "ExpectationsFlowsWritten61.4% (9,013)Dropped38.6% (5,678)ExpectationsAllFailures onlyNameActionFail %Failed recordsis_clean_contentDROP38.6%5678"

Run: Apr 10, 2026, 08:53 PM · () Completed

Run complete

Pipeline details

- Pipeline ID: 5e069f67-ee98-4f74-b2e9-4cf1edff...
- Pipeline type: ETL pipeline
- Creator: user_9501cd52
- Owner: user_9501cd52
- Run as: user_9501cd52
- Source code: 1 folder [Open in Editor](#)

Run details

- Pipeline run ID: c2d71cf3-9c34-4547-834b-c6b771...

Table metrics

Name	Catalog	Schema	Type	Dura...	Outp...	Expectations
gold_comments	iitb	bharat_bricks	Materialized...	5s	9K	1 unmet

gold_comments

Expectations

- Written: 61.4% (9,013)
- Dropped: 38.6% (5,678)

Expectations

Name	Action	Fail %	Failed records
is_clean_content	DROP	38.6%	5678

36 Click here.

The screenshot shows the Databricks Jobs & Pipelines interface. On the left is a sidebar with navigation options: Workspace, Recents, Catalog, Jobs & Pipelines (selected), Compute, Discover Beta, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, Playground, AI Gateway Beta, Experiments, Features, and Models.

The main area displays a pipeline named '02-data-transformation'. It shows a DAG with streaming tables 'silver_comments' and 'silver_posts' feeding into materialized views 'gold_comments' and 'gold_posts', which then feed into a 'gold_posts_chunked' table. The pipeline status is 'Completed' as of April 10, 2026, 08:53 PM.

On the right, the 'Pipeline details' section shows: Pipeline ID (5e069f67-ee98-4f74-b2e9-4cf1edff...), Pipeline type (ETL pipeline), Creator (user_9501cd52), Owner (user_9501cd52), Run as (user_9501cd52), and Source code (1 folder, Open in Editor).

Below the details is a table of pipeline runs. The first two rows are highlighted, showing 'gold_posts_chunked' with 979 output records.

Name	Catalog	Schema	Type	Dura...	Outp...	Expectations	Drop...	War...	Failed	Incremental	View in catalog
gold_posts_chunk...	iitb	bharat_bricks	Materialized...	5s	979	Not defined	-	-	-	Full recom...	View in catalog
gold_posts_chunk...	iitb	bharat_bricks	Materialized...	5s	979	Not defined	-	-	-	Full recom...	View in catalog

An orange circle highlights the 'View in catalog' link in the first row of the table.

37 Click "Sample Data"

The screenshot shows the Databricks Catalog Explorer interface. The left sidebar shows the 'Catalog' section with a tree view of the workspace, including 'workspace', 'system', 'iitb', 'bharat_bricks', and 'Tables (7)'. The 'gold_posts' table is selected.

The main area displays the 'gold_posts' table details. The 'Overview' tab is selected, showing a description: 'Clean r/iitbombay posts — filtered for deleted users, NSFW, profanity, and slurs content. Ready for dashboards, Genie, and RAG.'

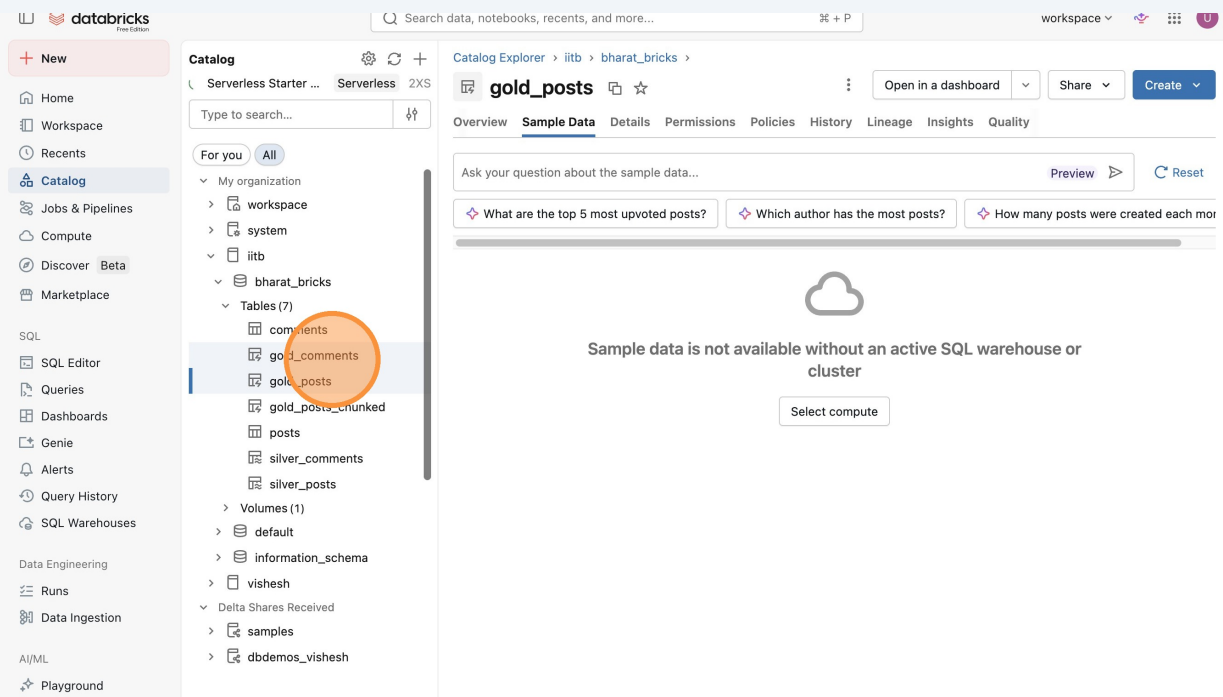
Below the description is a 'Compute not ready' message: 'To view the definition, select an active compute or wait for the current compute to be ready'. A 'Select compute' button is available.

Below the compute message is a table of columns. The first two columns are highlighted, showing 'post_id' and 'title'.

Column	Type	Comment	Tags	Column masking
post_id	string	Unique Reddit post identifier (e.g. 1id41gh). Primary key.		
title	string	Title/headline of the Reddit post as written by the author.		

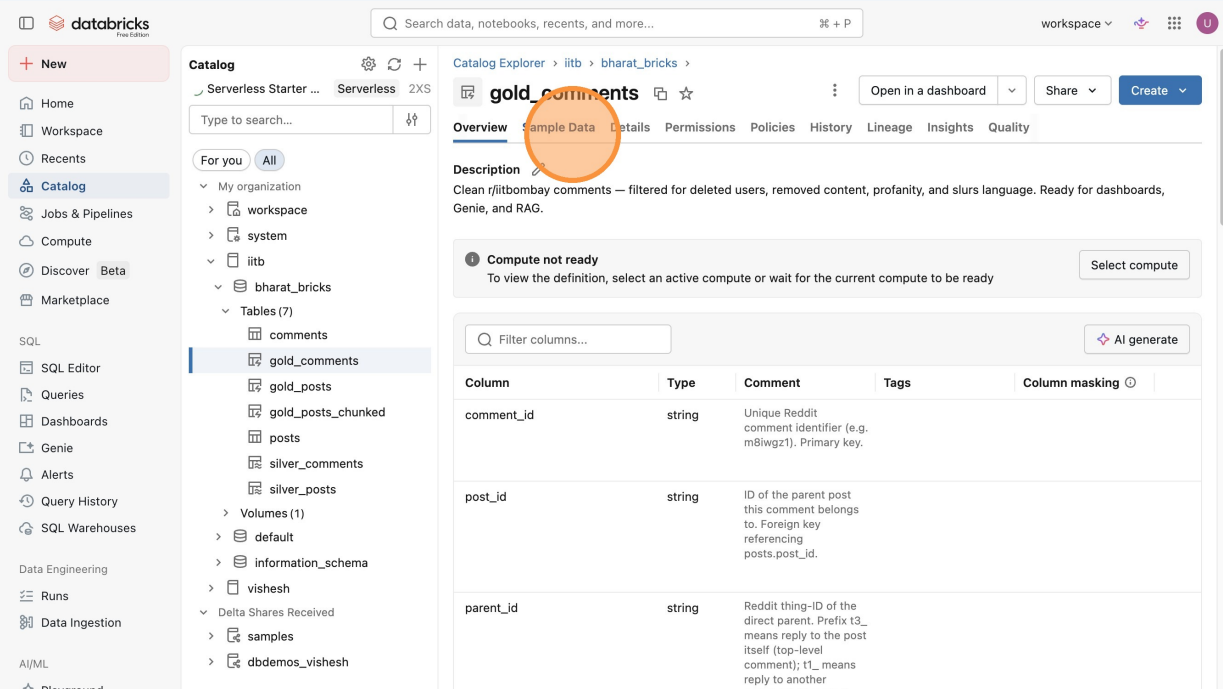
An orange circle highlights the 'Sample Data' tab in the top navigation bar.

38 Click "gold_comments"



The screenshot shows the Databricks Catalog Explorer interface. In the left sidebar, the 'Catalog' section is expanded, and the 'gold_comments' table is highlighted. The main panel displays the 'gold_posts' table, which is currently empty. A message states: "Sample data is not available without an active SQL warehouse or cluster". A "Select compute" button is visible below the message.

39 Click "Sample Data"



The screenshot shows the Databricks Catalog Explorer interface. In the left sidebar, the 'Catalog' section is expanded, and the 'gold_comments' table is highlighted. The main panel displays the 'gold_comments' table, which is currently empty. A message states: "Sample data is not available without an active SQL warehouse or cluster". A "Select compute" button is visible below the message.

The 'Sample Data' tab is highlighted in the top navigation bar. The 'Description' section shows: "Clean r/itbomabay comments — filtered for deleted users, removed content, profanity, and slurs language. Ready for dashboards, Genie, and RAG."

The 'Compute not ready' message states: "To view the definition, select an active compute or wait for the current compute to be ready".

The table structure is as follows:

Column	Type	Comment	Tags	Column masking
comment_id	string	Unique Reddit comment identifier (e.g. m8lwgzt1). Primary key.		
post_id	string	ID of the parent post this comment belongs to. Foreign key referencing posts.post_id.		
parent_id	string	Reddit thing-ID of the direct parent. Prefix t3_ means reply to the post itself (top-level comment); t1_ means reply to another		